Question Paper ID: 524841

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RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

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V Semester B.E. Regular / Supplementary Examination February/ March 2026

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Artificial Neural Networks and Deep Learning

Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

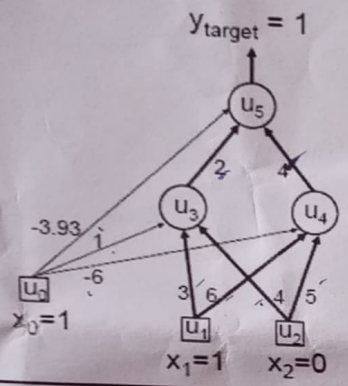
Part A

Question No	Question	M	CO	BT
1.1	Given η=0.1, error term (d-y) = 0.6, input x=[1, −2]. Compute Δw for LMS update.	02	3	3
1.2	A perceptron has w = [0.3, −0.2], b = 0.1, input x = [2, 1]. Compute net and output using step y = 1 if net > 0 else 0.	01	3	3
1.3	Differentiate (any 2 points) between learning with teacher and without teacher (focus on training signal + objective).	02	2	2
1.4	Perform max pooling on the given 4×4 feature map using a 2×2 window size with stride 2. Show the resulting pooled output.	02	1	3

3	1	1	3
2	5	0	2
1	4	2	1
4	7	2	4

1.5	Compute output size: Input 28×28, filter 3×3, padding = 1, stride = 1	02	1	3
1.6	What limitation of traditional neural networks led to the development of Recurrent Neural Networks?	01	1	1
1.7	What type of data is best suited for Recurrent Neural Networks?	01	1	1
1.8	What does a Bidirectional RNN consist of?	01	1	1
1.9	What is meant by exploitation in the Multi-Armed Bandit problem?	01	1	2
1.10	What is meant by exploration in the Multi-Armed Bandit problem?	01	1	2
1.11	Differentiate between on-policy and off-policy learning.	02	1	1
1.12	Differentiate conditional GAN from a standard GAN?	02	1	2
1.13	What happens when the Discriminator becomes too strong in a GAN?	02	1	2

Part B

Question No	Question	M	CO	BT
2	The network weights have been initialized as shown in Fig. Analyze the Multilayer perceptron Back Propagation algorithm for the initialized network by doing the following (use the sigmoid function): with a learning rate η = 0.25. a). Compute the net input and output of each hidden neuron and the final output neuron using the forward pass. b). Calculate the error term, the delta values for both output and hidden neurons, and determine the gradient direction. c). Update the weights using the backpropagation learning rule and show how the error decreases over two iterations of training. 	16	3	4
3a	Consider the convolutional neural network defined by the layers in the left column below. Fill in the shape of the output volume and the number of parameters at each layer. You can write the shapes in the NumPy format (e.g. (128,128,3)) (Perform step by step calculations) Notation: i). CONV5-N denotes a convolutional layer with N filters with height and width equal to 5. Padding is 2, and stride is 1. ii). POOL2 denotes a 2x2 max-pooling layer with stride of 2 and 0 padding. iii). FC-N denotes a fully connected layer with N neurons. Layer | Activation Volume Dimensions | Number of parameters Input | 32*32*1 | 0 CONV5-10 | | POOL2 | | CONV5-10 | | POOL2 | | FC10 | |	08	2	3
3b	Evaluate CNN features versus handcrafted features for CBIR.	08	4	4
	OR			
4a	Analyze how data augmentation improves generalization.	08	3	4
4b	Analyze how max pooling contributes to translation invariance and computational efficiency.	08	2	4

	Consider a simple Recurrent Neural Network (RNN) with the following setup: The RNN has one hidden layer with a hidden state vector ht which at each time step t is updated by the equation: $$h_t = \tanh(W_{hh}h_{t-1} + W_{xh}x_t + b_h)$$			
5a	Given the loss function $L = \sum_{t=1}^{T} \frac{1}{2}(y_t - \hat{y}_t)^2$ Compute the gradient of the loss with respect to Why and Whh (the output weight matrix) using BPTT.	08	3	3
5b	In a smart grid energy management system, an Echo State Network (ESN) is used to forecast energy demand based on real-time data from IoT sensors. The system receives input features such as current energy consumption, weather conditions, and time of day. The ESN is trained to predict energy demand for the next hour. Describe the architecture of the ESN in this scenario, highlighting the roles of the input, reservoir, and output layers.	08	3	3
	OR			
6a	Describe the training process of a basic RNN using Backpropagation Through Time (BPTT).	08	3	4
6b	Illustrate the internal working of an LSTM cell with gate-level operations and state transitions.	08	2	3
7a	Describe the complete architecture and working of a Reinforcement Learning system using an Agent–Environment model.	08	2	3
7b	With example briefly explain how neural network can be used function approximate in and RL setting such as playing Atari Games	08	2	3
	OR			
8a	Illustrate the working principle of Deep Q-Networks (DQN) in solving Atari games.	10	3	3
8b	Outline the role of Deep Learning in enhancing Reinforcement Learning systems.	06	2	2
9a	Explain Attention Mechanisms and their role in deep learning models.	10	3	2
9b	Describe the step-by-step training process of a Generative Adversarial Network (GAN) with a flow diagram.	06	2	2
	OR			
10a	Illustrate the complete architecture of a Neural Turing Machine with a neat diagram.	06	2	2
10b	Demonstrate how a Neural Turing Machine is superior to a standard neural network.	06	2	2
10c	Analyze the importance of external memory in a Neural Turing Machine.	04	1	1